

Evidential Matching, Uncertainty Calibration: Towards Robust Composed Video Retrieval with Noisy Triplets

Zhangtao Cheng

zhangtao.cheng@outlook.com

University of Electronic Science and Technology of China
Chengdu, Sichuan, China

Ting Zhong

zhongting@uestc.edu.cn

University of Electronic Science and Technology of China
Chengdu, Sichuan, China

Bozhu Zheng

bozhu.zheng@outlook.com

University of Electronic Science and Technology of China
Chengdu, Sichuan, China

Fan Zhou*

fan.zhou@uestc.edu.cn

University of Electronic Science and Technology of China
Chengdu, Sichuan, China
Key Laboratory of Intelligent Digital Media Technology of
Sichuan Province
Chengdu, Sichuan, China

Abstract

Composed video retrieval (CVR) aims to find a target video using a query formed by a reference video and a textual modification. Existing methods often presume that training triplets are correctly aligned; in practice, however, web-scale curation and annotator ambiguity lead to noisy triplet correspondence (NTC), where the modification may be incomplete (partial match) or incorrect (mismatch). We introduce **CURE**, an evidential **C**ross-modal **U**ncertainty **c**alib**R**ation fram**E**work that explicitly models and exploits alignment uncertainty to make CVR robust to NTC. CURE has three parts: (1) Uncertainty-guided Matching Filtration that models cross-modal uncertainty by parameterizing bidirectional evidence as a Dirichlet distribution and adaptively assigns weights to training triplets, generating fine-grained data partitions; (2) Robust Semantic Alignment, which rectifies the calibrated soft margins and relaxes the optimization from the hardest negatives to all negatives, thereby achieving noise-robust cross-modal alignment; and (3) Multi-level Relation Augmentation, which mines multi-level discriminative information via a semantic filtration strategy to facilitate modal alignment at both the video and caption levels, further alleviating the noise's adverse effects. Extensive experiments on benchmark datasets demonstrate the robustness and superiority of CURE, particularly in noisy scenarios.

CCS Concepts

• **Information systems** → **Retrieval tasks and goals**; • **Social and professional topics** → *User characteristics*.

Keywords

Composed Video Retrieval, Noisy Triplet Correspondence, Deep Evidential Learning, Cross-modal Alignment.

*Corresponding author.



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1 Introduction

With the rapid development of multimodal content on the internet, video has emerged as one of the most prevalent mediums for online information sharing and dissemination. Numerous video-centric platforms (e.g., TikTok and Snapchat) have proliferated, resulting in an exponential growth of video data. This explosion presents significant challenges for users seeking to obtain relevant videos that satisfy fine-grained requirements from large-scale video repositories. Such demands are often expressed through complex query compositions, motivating the task of *composed video retrieval* (CVR) [17, 36]. CVR aims to retrieve visually similar and semantically relevant target (*tar*) videos that reflect specific changes (e.g., altered objects, attributes, or actions) by querying with a reference (*ref*) video and a corresponding textual modification (*mod*). This task has numerous real-world applications, including e-commerce and fashion [15, 18], and internet-scale video search [8, 38], and multimodal question answering [5].

Recently, researchers have primarily addressed the CVR task by designing various multimodal architectures that unify visual and textual queries into a common representation aligned with target videos. These approaches can be broadly categorized into two groups: (1) *Fusion-based methods* [40] aim to jointly encode the reference video and textual instruction by applying a fusion strategy to combine their features into a unified representation, which is then used to compute similarity with candidate videos. (2) *Inversion-based methods* [35, 36] focus on learning a mapping network that projects the reference video into pseudo-textual tokens. These tokens are subsequently aligned with the target video features using well-labeled triplets in a self-supervised manner.

Despite significant progress, existing methods typically assume that all explicit triplets (i.e., $\langle ref, mod, tar \rangle$) are correctly aligned. In reality, however, this assumption is difficult to guarantee due

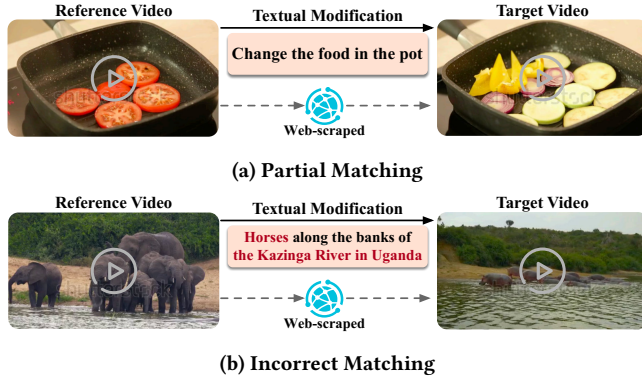


Figure 1: Illustration of noisy triplet correspondence on the WebCVR dataset. (a) Example of a partially matched triplet: the *mod* only partially describes the variation from *ref* to *tar*. (b) Example of a completely mismatched triplet: the *mod* fails to describe any relevant features of the *tar*.

to several inevitable sources of noise. First, textual ambiguity remains a persistent challenge: modification texts are often vague or incomplete (see Figure 1 (a)), thereby leading to partially matched triplets [41]. Second, to reduce annotation costs, many datasets are constructed using web crawling or large language models [17, 36], which inevitably introduce noise into cross-modal correspondences due to annotator imperfections, resulting in incorrect matching issues (see Figure 1 (b)). Based on these observations, we reveal and study a new problem in the CVR task: *noisy triplet correspondence* (NTC) [14, 16]. Specifically, NTC refers to false correspondences (i.e., mismatched alignments) within annotated triplets. NTC arises when negative triplets are mistakenly treated as positive supervision during cross-modal learning, thereby misleading the model to overfit noisy signals and converge to suboptimal solutions [2].

To address this problem, a straightforward solution is to directly use existing robust cross-modal methods [16, 27] to mitigate the adverse effects of NTC. However, directly transferring these methods to the CVR task faces several unresolved challenges:

Challenge 1: Absence of triplet correspondence. Prior studies primarily address noisy correspondence in image-text pairs, without a robust design to handle the complex noise in triplets, leading to suboptimal CVR performance. Specifically, in Figure 1, the *mod* may only partially describe the changes from *ref* to *tar*, yielding partially matched triplets (i.e., matched modification-target pairs). Moreover, *mod* may incorrectly specify attributes absent in *tar*, resulting in completely mismatched triplets. Thus, learning with NTC is inherently more challenging and complex, requiring handling both matched pairs and mismatched triplets.

Challenge 2: Overlooking cross-modal matching uncertainty. Existing methods typically adopt co-network techniques to identify noisy samples based on predicted per-sample loss values [16] or similarity scores [39]. However, these methods often suffer from cross-modal uncertainty arising from NTC in the CVR task, which ultimately leads to incorrect data partitioning: truly mismatched pairs may be mistakenly classified as matched, and vice versa [22]. Importantly, alignment uncertainty can be construed as a penalty on the retrieved outcomes, reflecting the degree to which texts should be discouraged from aligning with a particular visual identity.

To this end, we propose **CURE**, a novel evidential Cross-modal Uncertainty calibration framEwork for enhancing robust CVR by quantifying the uncertainty arising from noisy triplet correspondence. Specifically, inspired by evidential deep learning (EDL) [20], we design an *uncertainty-guided matching filtration* (UMF) to learn bidirectional cross-modal evidence via regarding the CVR task as K -way classification task, where a query is classified based on its corresponding match in another modality. Then, UMF captures the alignment uncertainty caused by NTC via parameterizing the bidirectional evidence into Dirichlet distribution based on the cross-modal similarity. UMF utilizes the quantized uncertainty to adaptively assign weights to each training triplets, which is exploited to dynamically separate the data into fine-grained sets with clean, noisy and hard-determined triplets.

Second, we propose a *Robust Semantic Alignment* (RSA) module to achieve noise-robust cross-modal alignment by providing a reliable recasting of soft margins through the modeled uncertainty, calibrated using confidence-aware yet error soft correspondence labels. RSA further relaxes similarity learning from only the hardest negative samples to all negative samples by introducing an upper bound constraint. This design leads to more stable optimization under NTC scenarios, while still benefiting from hard negative mining to facilitate effective cross-modal alignment. Finally, we design a *Multi-level Relation Augmentation* (MRA) mechanism to further mitigate the adverse effects of partial matching within the clean set. MRA selects multi-level discriminative information through a semantic filtration strategy and reconciles the multi-level composed query-video similarity distributions for proper cross-modal alignments at both video and caption levels. Notably, although these three modules primarily address the NTC problem, they remain effective even on clean triplets, as visual variations are often too complex to be fully expressed by text, resulting in implicit NTC.

In summary, the key contributions of our work are as follows:

- We are the first to explore the problem of noisy triplet correspondence in the CVR task. To address this challenge, we propose a novel evidential cross-modal uncertainty calibration framework, CURE, designed to enhance robust CVR by explicitly quantifying the cross-modal uncertainty arising from NTC.
- The UMF module is to capture cross-modal uncertainty by parameterizing evidence as a Dirichlet distribution and generate fine-grained data partitions. The RSA module rectifies the calibrated soft margins and performs optimization over all negative samples to achieve noise-robust alignment. The MRA module further mitigates the noise’s effects by mining discriminative cross-modal semantics, facilitating multi-level alignment.
- We conduct extensive experiments on two real-world datasets to validate the effectiveness of CURE. The results demonstrate that CURE advances the state-of-the-art CVR, enhancing robustness against NTC, particularly under high noise rates. The code and datasets to reproduce the results are available at <https://github.com/zbzzbzzb/CURE>.

2 Related Work

Composed Video Retrieval. With the rapid development of multimodal content on the internet, *composed image retrieval* (CIR)

[7, 12, 21, 37] has emerged as a prominent task, wherein a user provides a reference image along with a textual description specifying the desired modification. Existing CIR approaches [4, 10, 31] typically leverage pre-trained vision-language models (e.g., BLIP [23] and CLIP [30]) to learn joint multimodal representations that capture both the reference image and the relative textual modification for retrieving the target image. However, due to the absence of temporal dynamics, CIR methods are inherently limited to static content modifications and cannot be directly extended to the CVR task. Recently, researchers [17, 36] have extended the CIR task from the image domain to the video domain by formulating the problem of CVR, where the goal is to retrieve a target video given a reference video and a corresponding compositional change description. Specifically, CoVR [36] leverages BLIP to perform cross-modal alignment and fusion for retrieval. EgoCVR [17] proposes a training-free re-ranking strategy to improve retrieval performance. FDCA [40] enhances fine-grained alignment by decomposing textual features into three semantically meaningful components.

However, existing works typically assume that all explicit triplets are correctly aligned, making them ill-suited to address the problem of NTC. To overcome this limitation, we propose an evidential cross-modal uncertainty calibration framework that models cross-modal uncertainty to enhance the robustness in the CVR task.

Noisy Correspondence Learning. As a specialized learning paradigm under noisy supervision [25], *noisy correspondence* (NC) [16] refers to alignment errors in multimodal learning scenarios. A common manifestation of NC is the *false positive* problem, where semantically irrelevant multimodal pairs are incorrectly treated as matched. To mitigate this issue, a growing body of work has proposed techniques applicable across various downstream tasks, including cross-modal retrieval, visible-infrared person re-identification, visual question answering, and image captioning. NCR [16] was among the first to formally define the NC problem and introduced a co-divide mechanism that leverages the memorization effect of neural networks to separate clean and noisy samples. Building upon this, subsequent works [11, 39] focus on improving label reliability through refined estimation strategies to enhance model robustness. Other approaches aim to design noise-resistant loss functions, such as evidential loss [26, 28, 42], complementary contrastive loss [29], and active complementary loss [29]. Notably, we are the first to study the problem of NTC and to introduce evidential deep learning for the CVR task, enhancing robustness by modeling cross-modal alignment uncertainty.

3 Methodology

Problem Definition: Composed Video Retrieval. Let $\mathcal{V}$ denote the set of videos and $\mathcal{T}$ the set of textual instructions. Given a query $q = (q_v, q_t)$, where $q_v \in \mathcal{V}$ is a reference video and $q_t \in \mathcal{T}$ is a textual modification, the objective of the CVR task is to retrieve a target video $v \in \mathcal{D}$ from a large-scale video database $\mathcal{D} = \{v_1, \dots, v_N\}$ that preserves the visual content of q_v while incorporating the intended modifications described by q_t . Formally, the task can be defined via a scoring function $\omega : \mathcal{V} \times \mathcal{T} \times \mathcal{V} \rightarrow \mathbb{R}$, which measures the semantic similarity between the reference video q_v , the modification text q_t , and each candidate video $v_i \in \mathcal{D}$. The video with the highest score under ω is selected as the retrieval result.

The CVR task is particularly challenging as it requires bridging the domain gap between the visual and textual modalities.

Figure 2 illustrates the key components and their relationships of our CURE framework. The following sections delve into the specifics of each component and their respective implementations.

3.1 Multi-modal Feature Extraction

3.1.1 Text Feature Extraction. In the text processing branch, inspired by existing CVR methods [35], we first utilize a language-for-vision strategy that transforms visual signals into linguistic form, providing a rudimentary semantic understanding of the reference video q_v . Specifically, we employ a pre-trained captioning model Ψ_c (e.g., BLIP [23]) to generate a textual description of the reference video: $q_c = \Psi_c(q_v)$. The generated caption q_c resides in the natural language domain, making it more interpretable and compatible with downstream text-based reasoning. Furthermore, we leverage the text encoder Ψ_t of vision-language models (VLMs), such as BLIP, to encode the instruction text q_t and obtain its textual embedding: $\mathbf{X}^t = \Psi_t(q_t) \in \mathbb{R}^d$. Analogously, the caption representation $\mathbf{X}^c$ is derived using the same encoder: $\mathbf{X}^c = \Psi_t(q_c)$.

3.1.2 Vision Feature Extraction. For video content, we utilize the visual encoder Ψ_v from the VLM to model the sampled reference video frames $\mathcal{F} = \{v_1, v_2, \dots, v_F\}$, where F denotes the number of sampled frames. Each frame v_i is first divided into m non-overlapping patches, which are projected into a sequence of patch tokens $\mathbf{E}_i \in \mathbb{R}^{m \times d_v}$. These tokens are then passed through the visual encoder Ψ_v , yielding the frame-level visual embedding: $\mathbf{f}_i^v = \Psi_v(\mathbf{E}_i) \in \mathbb{R}^d$. Furthermore, we adopt a non-parametric strategy by averaging the embeddings of all frames $\mathbf{F}^v = \{\mathbf{f}_1^v, \mathbf{f}_2^v, \dots, \mathbf{f}_F^v\}$ to obtain the final video representation: $\mathbf{X}^v \in \mathbb{R}^d$.

3.1.3 Cross-modal Fusion. To extract the retained components of the modification associated with the reference video, we design a cross-attention strategy to fuse critical semantic information from the instruction q_t and the video q_v . Specifically, we project the visual feature $\mathbf{X}^v$ and textual feature $\mathbf{X}^t$ into a shared latent space to compute the query, key, and value representations: $\mathbf{Q} = \phi_Q(\mathbf{X}^v)$, $\mathbf{K} = \phi_K(\mathbf{X}^t)$, $\mathbf{V} = \phi_V(\mathbf{X}^t)$, where ϕ_Q , ϕ_K , and ϕ_V are learnable linear projection layers. The retained textual feature guided by the reference video is computed via reference-guided cross-attention: $\mathbf{X}^{tv} = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right)\mathbf{V} + \mathbf{V}$. Analogously, to capture the interaction between the reference video's caption and the modification, we compute the fused features $\mathbf{X}^{tc}$ using the same procedure. Finally, we adopt a non-parametric aggregation strategy by averaging the learned embeddings to form the final query representation $\mathbf{X}^q$.

3.2 Uncertainty-guided Matching Filtration

To effectively mitigate the impact of uncertainty arising from unreliable matching triplets (i.e., noisy triplet correspondence) in the CVR task, it is essential to explicitly model cross-modal alignment uncertainty. Inspired by the EDL paradigm [1, 32], we propose an uncertainty-guided matching filtration (UMF) to identify noisy triplets during training by modeling alignment uncertainty. Specifically, we first extract the evidence from each cross-modal alignment and use it to construct a prior Dirichlet distribution over the matching probabilities. Following the principles of Subjective Logic [20],

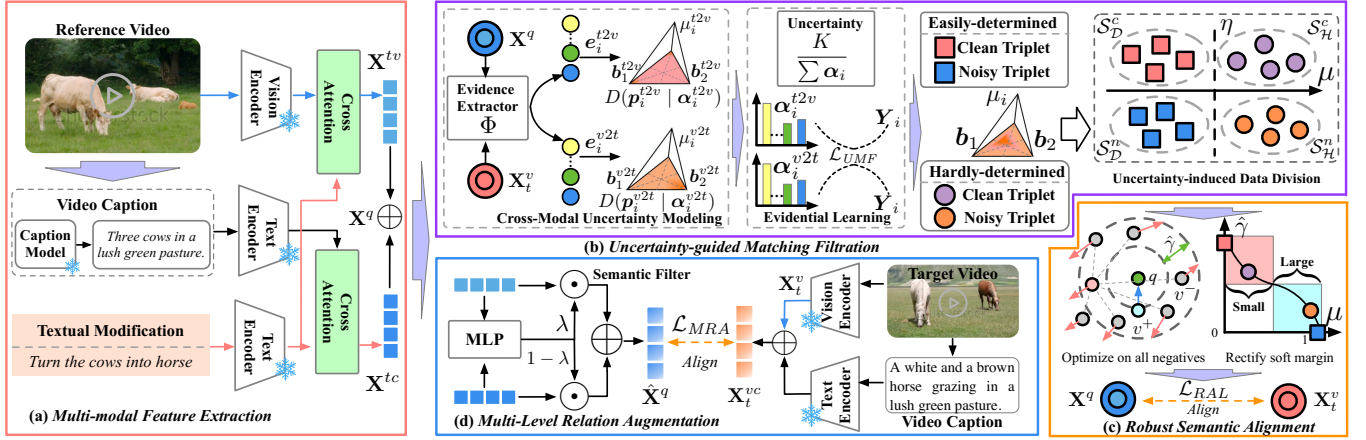


Figure 2: Overall framework of CURE. (a) The detail process of multi-modal feature extraction for the composed query. (b) Uncertainty-guided matching filtration learns uncertainty by parameterizing bidirectional evidence and adaptively identifies noisy instances. (c) Robust semantic alignment rectifies the calibrated soft margins and relaxes optimization over all negative pairs. (d) Multi-Level relation augmentation mines discriminative semantic information at both the video and caption levels to mitigate the effects of noise through multi-level alignment.

we then quantify the alignment uncertainty, which serves as a criterion to guide the division between clean and noisy triplets.

3.2.1 Cross-Modal Uncertainty Modeling. Given a query triplet (q_t, q_v, v) , we adopt a pseudo-textual strategy [10] that combines the features of the reference video q_v and the textual modification q_t into a unified pseudo-text representation $\hat{t}$. This design reformulates the original triplet-based alignment into a text-to-video matching problem, thereby simplifying the modeling of cross-modal uncertainty in CVR and avoiding the introduction of additional model parameters. For ease of representation, we focus on computation within a mini-batch containing K triplets. Specifically, for a given pair $(\hat{t}_i, v_j)$, we extract the alignment evidence using an evidence extractor Φ , defined as:

$$e_{ij} = \Phi(\hat{t}_i, v_j) = \exp^{f(S(\hat{t}_i, v_j))} = \exp^{f(S(X_i^q, X_j^v))}, \quad (1)$$

where f represents the combination of ReLU and Tanh function. X_j^v denotes the target video v_j 's representations. Following Eq. 1, we obtain the textual query's evidence vector as $e_i^{t2v} = [e_{i1}, \dots, e_{iK}]$ based on the corresponding cross-modal similarity. Analogously, for a visual query v_i , the evidence vector is computed as $e_i^{v2t} = [e_{i1}, \dots, e_{iK}]$ using the same process.

Through Subjective Logic, CURE assigns a belief probability b_{ij} to each query and calculates the uncertainty u_i of overall alignment based on the cross-modal evidence (i.e., e_i^{t2v} and e_i^{v2t}) as follows:

$$b_{ij} = \frac{e_{ij}}{\beta_i} = \frac{\alpha_{ij} - 1}{\beta_i}, \quad u_i = 1 - \sum_{j=1}^K b_{ij} = \frac{K}{\beta_i}, \quad (2)$$

where $\beta_i = \sum_{j=1}^K (e_{ij} + 1)$ represents the intensity of Dirichlet distribution strength. The belief probability $b_i = \{b_{ij}\}_{j=1}^K$ is regarded as the subjective opinion, derived from the parameters of the corresponding Dirichlet distribution $\alpha_i = \{\alpha_{ij}\}_{j=1}^K$.

Intuitively, we formulate the CVR task as its K -way classification problem, where a query is classified into its correct cross-modal

counterpart. By parameterizing a Dirichlet distribution with evidence, we can obtain the density of each probability assignment, enabling the modeling of second-order probabilities and cross-modal alignment uncertainty. Essentially, the Dirichlet distribution serves as a probability density function that is used to calculate possible values of the alignment probability. The density function is characterized by $\alpha_i = \{\alpha_{ij}\}_{j=1}^K$ and is defined as:

$$D(\mathbf{p}_i | \alpha_i) = \begin{cases} \frac{1}{B(\alpha_i)} \prod_{j=1}^K p_{ij}^{\alpha_{ij}-1}, & \text{for } \mathbf{p}_i \in \mathcal{X}_K, \\ 0, & \text{otherwise,} \end{cases} \quad (3)$$

where $B(\alpha_i)$ denotes the K -dimensional multinomial Beta function, and $\mathbf{p}_i \in \mathbb{R}^K$ represents the cross-modal alignment probabilities. $\mathcal{X}_K$ refers to the K -dimensional unit simplex [13].

3.2.2 Bidirectional Evidential Learning. First, we compute the bidirectional cross-modal evidence using Eq. 1, and then utilize a least-squares loss to assign proper on each retrieval, thereby making the class probabilities $\mathbf{p}_i$ to approximate the ground-truth Y_i , i.e., its cross-modal counterpart. Here, $Y_i \in \mathbb{R}^K$ is a K -dimensional one-hot vector with the i -th element set to 1 and all others set to 0. Notably, the density of $\mathbf{p}_i$ is based on the parameterized Dirichlet distribution α_i . Thus, the least-squares loss function $\mathcal{L}_e(\alpha_i, Y_i)$ is summarized as follows:

$$\begin{aligned} \mathcal{L}_e(\alpha_i, Y_i) &= \int \|Y_i - \mathbf{p}_i\|_2^2 \frac{1}{B(\alpha_i)} \prod_{j=1}^K p_{ij}^{\alpha_{ij}-1} d\mathbf{p}_i \\ &= \sum_{j=1}^K \left(Y_{ij} - \frac{\alpha_{ij}}{\beta_i} \right)^2 + \frac{\alpha_{ij}(\beta_i - \alpha_{ij})}{\beta_i^2(\beta_i + 1)}, \end{aligned} \quad (4)$$

where the first term is the squared error $(Y_{ij} - \mathbb{E}(p_{ij}))^2$, and the second term represents the variance $\text{Var}(p_{ij})$. The expectation and variance of p_{ij} are denoted by $\mathbb{E}(p_{ij})$ and $\text{Var}(p_{ij})$, respectively. The expected probability is estimated by $\frac{\alpha_{ij}}{\beta_i}$ [32]. By minimizing the loss $\mathcal{L}_e$, CURE is encouraged to assign higher evidence

to matched (positive) triplets compared to mismatched (negative) ones. However, it remains challenging to generate scant evidence for mismatched triplets. Ideally, the generated evidence of negative samples is expected to trend towards zero in this case. To achieve this, we introduce a Kullback–Leibler (KL) divergence regularization term that penalizes deviations from the uncertain retrieval:

$$\begin{aligned}\mathcal{L}_{kl}(\alpha_i, Y_i) &= KL[D(\mathbf{p}_i | \tilde{\alpha}_i) \| D(\mathbf{p}_i | \mathbf{1})] \\ &= \log \left(\frac{\Gamma(\sum_{j=1}^K \tilde{\alpha}_{ij})}{\Gamma(K) \prod_{j=1}^K \Gamma(\tilde{\alpha}_{ij})} \right) + \sum_{j=1}^K (\tilde{\alpha}_{ij} - 1) \left[\psi(\tilde{\alpha}_{ij}) - \psi\left(\sum_{j=1}^K \tilde{\alpha}_{ij}\right) \right],\end{aligned}\quad (5)$$

where Γ and ψ denotes the gamma and digamma functions, respectively. The Dirichlet parameter $\tilde{\alpha}_i = Y_i + (1 - Y_i) \odot \alpha_i$ is calculated by eliminating the unreliable evidence from the predicted Dirichlet distribution α_i . Once we obtain the evidence vector $\mathbf{e}_i^{t2v} \in \mathbb{R}^K$, the corresponding Dirichlet distribution could be calculated as $\alpha_i^{t2v} = \mathbf{e}_i^{t2v} + 1$. Thus, the textual query evidential loss is:

$$\mathcal{L}^{t2v}(\hat{t}_i, \tilde{y}_i) = \mathcal{L}_e(\alpha_i^{t2v}, \tilde{y}_i) + \xi_2 \mathcal{L}_{kl}(\alpha_i^{t2v}, \tilde{y}_i), \quad (6)$$

where $0 < \xi_2 < 1$ denotes a balance factor. The soft correspondence label is assigned as $\tilde{y}_i = 1$ if $i = \arg \max(\mathbf{e}_i^{t2v} + \mathbf{e}_i^{v2t})$, and $\tilde{y}_i = 0$ otherwise. Analogously, the evidential loss $\mathcal{L}_e^{v2t}(v_i, \tilde{y}_i)$ for video-to-text retrieval is computed using the same procedure. Finally, the overall bidirectional evidential loss is defined as: $\mathcal{L}_{UMF} = \mathcal{L}^{t2v}(\hat{t}_i, \tilde{y}_i) + \mathcal{L}^{v2t}(v_i, \tilde{y}_i)$.

3.2.3 Uncertainty-induced Data Division. After computing the bidirectional uncertainty and averaging it to obtain the overall uncertainty u , we design an uncertainty-induced division strategy to construct a reliable and fine-grained data partition. Specifically, we divide the training data into two subsets: a determined set $\mathcal{S}_{\mathcal{D}}$ and a hard set $\mathcal{S}_{\mathcal{H}}$, using a threshold η on the uncertainty score u . The first set $\mathcal{S}_{\mathcal{D}}$ is composed of the data triplet with definitive one-hot labels wherein clean data labeled as $\tilde{y} = 1$ and noisy data labeled as $\tilde{y} = 0$. In contrast, the hard set $\mathcal{S}_{\mathcal{H}}$ consists of triplets where the labels are discarded and replaced with soft correspondence scores $\tilde{y} \in [0, 1]$, reflecting the estimated degree of alignment. The division procedure can be summarized as:

$$\begin{cases} \mathcal{S}_{\mathcal{D}} \supset (\hat{t}_i, v_j), & u_i < \eta \\ \mathcal{S}_{\mathcal{H}} \supset (\hat{t}_i, v_j), & u_i \geq \eta \end{cases} \quad (7)$$

For the determined set $\mathcal{S}_{\mathcal{D}}$, a soft correspondence label $\tilde{y}_i$ is assigned to each pair. Specifically, $\tilde{y}_i = 1$ indicates that the pair $(\hat{t}_i, v_j)$ belongs to the clean subset $\mathcal{S}_{\mathcal{D}}^c$, while $\tilde{y}_i = 0$ assigns it to the noisy subset $\mathcal{S}_{\mathcal{D}}^n$. For the hard subset $\mathcal{S}_{\mathcal{H}}$, the label $\tilde{y}_i$ is also used to determine its partition. $\mathcal{S}_{\mathcal{H}}$ is further divided into two subsets: a hardly-determined clean subset $\mathcal{S}_{\mathcal{H}}^c$ and a hardly-determined noisy subset $\mathcal{S}_{\mathcal{H}}^n$. The value of $\tilde{y}_i$ determines whether CURE performs positive or negative learning for the corresponding triplet.

3.3 Robust Semantic Alignment

Existing works typically utilize the Triplet Ranking Loss (TRL) [9] for cross-modal alignment in the CVR task, achieving promising performance by exploiting the hardest negatives. However, this

strategy degrades severely in the presence of noisy triplet correspondence, as the hardest negatives tend to amplify erroneous gradients caused by the noisy correspondence [14]. Moreover, under our explicit partitioning, we observe two embarrassing phenomenons in the hard set $\mathcal{S}_{\mathcal{H}}$: truly matched triplets are mistakenly treated as mismatched and vice versa. To address these limitations, we propose a Robust Alignment Loss (RAL) to enhance the robustness of models against noise in the CVR task. RAL rectifies the calibrated soft margins $\hat{y}$ and further mitigates the influence of noise by relaxing the optimization objective from the hardest negatives to all negatives, constrained by an upper margin. Formally, for an input pair $(\hat{t}_i, v_i)$, RAL is defined as:

$$\begin{aligned}\mathcal{L}_{RAL}(\hat{t}_i, v_i) &= \left[\hat{y} - S_{t2v}^+(\hat{t}_i) + \tau \log \left(\sum_{j=1}^K \exp(S(\hat{t}_i, v_j)/\tau) \right) \right]_+ \\ &\quad + \left[\hat{y} - S_{v2t}^+(v_i) + \tau \log \left(\sum_{j=1}^K \exp(S(\hat{t}_j, v_i)/\tau) \right) \right]_+, \end{aligned} \quad (8)$$

where τ represents a temperature coefficient to control hardness and K is the batch-size. $[x]_+ \equiv 0$. The trusted soft margin $\hat{y}$ could be adaptively determined by:

$$\hat{y}_i = \begin{cases} \frac{m^{\tilde{y}_i} - 1}{m - 1} \gamma_i, & (\hat{t}_i, v_i) \in \mathcal{S}_{\mathcal{D}}, \\ \frac{m^{\tilde{y}_i} - m^{(2u_i - 1)\tilde{y}_i}}{m - 1} \gamma_i, & (\hat{t}_i, v_i) \in \mathcal{S}_{\mathcal{H}}, \end{cases} \quad (9)$$

where γ is a positive margin coefficient. Theoretically, the learned soft margins $\hat{y}_i$ are expected to be higher for matched data pairs with low uncertainty and lower for mismatched pairs with high uncertainty, while remaining bounded above by the initial margin γ . This ensures that all pairs in the hard partition contribute less to the loss than those in the easily-determined clean partition. In addition, RAL reduces the risk of optimization being dominated by the hardest negatives, thereby stabilizing the training process and making it more comprehensive by considering all negative pairs.

3.4 Multi-Level Relation Augmentation

Due to limited attention spans and domain expertise, annotators may partially describe the attribute changes between the reference and target videos, rendering the modification q_t a weak caption for retrieving the target video v . To further mitigate overfitting caused by partial matching, we propose a multi-level relation augmentation (MRA) mechanism that mines discriminative semantics to enable multi-level cross-modal alignment. Specifically, we first design a semantic filter that selects representative semantic information and adaptively fuses the features of two unified queries, $\mathbf{X}^{t^v}$ and $\mathbf{X}^{t^c}$, via a learnable linear weighted addition. The fusion weights are dynamically computed based on the input context. This design enables the model to adaptively extract task-relevant semantics from both types of unified queries, thereby improving its flexibility in handling diverse CVR scenarios with varying modification demands. Importantly, the linear fusion operation is specifically designed to maintain the fused query embedding still residing in the VLM's original embedding space, where the target video embedding also resides, thereby facilitating effective target retrieval. The semantic

Table 1: Performance comparison on two real-world datasets under varying noise rates. The best results are in bold font and the second underlined. Lower values of MnR, and higher values of R@k, indicate better performance.

Dataset	Noise	Metric	BLIP	CLIP	CoVR	CVRDE	CVRDM	FDCA	Pic2word	I2W	LinCIR	Uncer-R	NCR	RCL	TME	CURE	Improv.
WebCVR	30%	R@1	14.47	16.90	34.87	<u>36.37</u>	33.53	23.70	19.33	18.04	20.73	12.63	9.77	15.37	31.65	41.27	13.47%
		R@5	34.07	39.57	60.43	<u>62.33</u>	58.53	49.73	34.00	33.91	36.60	24.20	22.87	33.60	55.41	69.37	11.29%
		R@10	44.60	50.47	70.43	<u>71.83</u>	67.70	60.73	40.10	41.15	42.80	31.00	31.17	43.27	64.85	78.47	9.24%
		MnR	89.43	66.26	41.89	<u>31.47</u>	65.14	71.14	156.96	195.13	107.99	199.45	155.69	125.02	99.39	23.80	24.37%
	60%	R@1	7.17	12.10	32.37	<u>35.50</u>	31.80	18.73	10.90	9.77	11.93	9.93	7.70	11.67	31.18	40.39	13.77%
		R@5	21.47	32.70	57.40	<u>60.00</u>	55.53	43.33	19.40	19.24	21.10	20.00	19.03	28.03	55.85	68.29	13.81%
		R@10	30.60	44.43	67.57	<u>70.23</u>	66.30	54.73	23.13	23.47	24.77	25.33	26.27	37.53	66.06	77.79	10.76%
		MnR	131.82	84.19	55.72	<u>39.92</u>	75.35	97.72	177.08	238.14	152.35	282.28	220.52	146.03	136.45	27.89	30.13%
	90%	R@1	0.40	2.57	27.83	30.67	27.17	0.47	2.80	2.73	2.77	7.10	1.67	7.23	<u>31.01</u>	34.90	12.21%
		R@5	1.37	11.20	50.27	53.70	49.70	5.13	4.90	5.07	5.33	13.33	5.13	19.93	<u>54.27</u>	60.23	10.98%
		R@10	17.80	18.07	60.67	<u>63.33</u>	59.17	517.19	5.80	6.34	6.47	16.47	9.20	27.50	<u>62.20</u>	71.23	12.47%
		MnR	768.03	203.21	100.80	<u>65.46</u>	112.10	249.0	358.69	460.22	461.54	580.91	363.43	271.82	287.16	43.57	33.44%
FineCVR	30%	R@1	15.65	16.69	31.95	31.13	<u>32.28</u>	18.00	5.74	5.06	6.47	4.42	3.76	6.49	27.54	34.76	7.68%
		R@5	40.65	43.51	65.02	64.27	<u>65.52</u>	44.81	16.26	14.96	18.87	14.93	14.33	20.84	58.70	68.59	4.96%
		R@10	54.12	57.56	78.25	77.99	<u>78.50</u>	59.93	22.80	21.20	26.73	22.55	22.58	29.98	72.32	81.33	3.61%
		MnR	46.90	34.56	14.83	<u>13.49</u>	15.97	41.61	253.39	237.54	275.03	267.47	255.53	215.96	32.72	11.30	16.23%
	60%	R@1	9.43	12.76	29.49	29.12	<u>30.33</u>	13.41	2.97	2.84	3.65	3.05	2.44	4.98	27.11	32.95	8.63%
		R@5	27.47	36.11	61.88	61.38	<u>62.16</u>	37.20	8.74	8.39	10.84	10.94	11.82	16.94	57.86	66.53	7.03%
		R@10	39.47	50.05	75.08	75.55	<u>76.01</u>	51.50	12.65	12.13	14.97	17.43	18.53	25.20	70.80	79.78	4.95%
		MnR	88.61	51.89	21.64	<u>17.81</u>	21.74	63.02	496.61	336.85	447.92	307.46	321.3	273.41	39.41	12.84	27.90%
	90%	R@1	0.44	4.05	23.41	23.88	23.59	1.61	0.82	0.80	0.94	0.86	1.60	3.77	<u>24.35</u>	26.78	9.98%
		R@5	1.85	14.24	51.71	52.61	52.00	6.89	2.12	2.12	2.53	3.99	8.22	12.24	<u>53.77</u>	58.25	8.33%
		R@10	3.22	22.66	66.03	66.77	65.96	11.94	3.10	3.24	3.85	6.69	13.65	19.55	<u>66.97</u>	72.17	7.76%
		MnR	1030.15	197.21	53.75	<u>38.85</u>	58.23	453.83	1120.32	1156.86	1060.92	1178.12	405.8	381.15	74.51	21.87	43.71%

filtering process is formally defined as:

$$\hat{X}^q = \lambda * X^{tv} + (1 - \lambda) * X^{tc}, \quad \lambda = \sigma(\text{MLP}([X^{tv} \parallel X^{tc}])), \quad (10)$$

where f_q is the final composed query feature, λ is the trade-off weight for balancing the importance of the unified textual query and unified visual query. $\parallel$ is the feature concatenation operation, MLP refers to the multi-layer perceptron, and σ denotes the Sigmoid function. Furthermore, we generate the target video's caption using the captioning model Ψ_c and obtain the corresponding representation X_t^c . We then sum the visual and caption features of the target video to form a vision-caption complementary feature: $X_t^{vc} = X_t^v + X_t^c$. Finally, we adopt contrastive learning to align the query representation with the target video at both the visual and caption levels, thereby achieving multi-level semantic alignment:

$$\mathcal{L}_{MRA} = -\frac{1}{\sum_i^K \tilde{y}_i} \sum_{i,j \neq i}^K \tilde{y}_i \log \left(1 - \frac{\exp(\tau(\hat{X}_i^q)^T X_{t,i}^{vc})}{\sum_j^K \exp(\tau(\hat{X}_j^q)^T X_{t,j}^{vc})} \right). \quad (11)$$

Notably, the semantic filter acts as a textual refiner that enhances the textual modification q_t , thereby improving text-video alignment in the CVR task. This augmentation strengthens the expressiveness of the modification and mitigates overfitting caused by partially mismatched triplets in the clean set $\mathcal{S}^c$.

3.5 Video Retrieval

To accomplish the CVR task, we leverage the advanced capabilities of the VLM (i.e., BLIP), as a retriever. Specifically, after obtaining the fused query representation X^q , we employ the visual encoder $\Psi_v(\cdot)$ from the VLM to encode each candidate video $v \in \mathcal{D}$ from the

video database $\mathcal{D}$. Thus, the target video v_t is retrieved as follows:

$$v_t = \operatorname{argmax}_{v \in \mathcal{D}} \frac{\Psi_v(v)^T X^q}{\|\Psi_v(v)\| \cdot \|\Psi_t(X^q)\|}. \quad (12)$$

where the final retrieved target video v_t corresponds to the candidate video that is most similar to the composed query q .

The training of our CURE is guided by three loss functions: $\mathcal{L}_{UMF}$, $\mathcal{L}_{RAL}$, and $\mathcal{L}_{TRA}$. The overall training objective is jointly optimized upon three loss functions: $\mathcal{L} = \mathcal{L}_{RAL} + \rho \mathcal{L}_{UMF} + \varsigma \mathcal{L}_{MRA}$, where λ and ς are hyperparameters that balance the contributions of the evidential and relation augmentation losses.

4 Experiments

4.1 Experimental Settings

A concise summary of the experimental settings is provided below, with further details available in Appendix A.

- *Datasets.* For evaluation, we use two benchmark datasets following prior works [36, 40]: **WebCVR** [36] and **FineCVR** [40]. In addition, we introduce a new dataset, **WebCVR-8M** [36], to further assess the generalizability of our proposed CURE.

- *Baselines.* To evaluate the effectiveness of our CURE, we compare CURE with 13 competitive baselines divided into three groups: (1) *Fusion-based methods*: CLIP [30], BLIP [23], CoVR [36], FDCA [40], CVRDE [35] and CVRDM [34]. (2) *Inversion-based methods*: Pic2word [31], I2W [33], and LinCIR [10]. (3) *Robust methods*: Uncer-R [6], NCR [16], RCL [14], and TME [24].

- *Metrics.* Following prior work [19, 36], we adopt widely used retrieval metrics to evaluate performance, including Recall at rank K (R@K) and Mean Rank (MnR).

4.2 Main Results

Table 1 presents the comparative results between CURE and 13 competitive baselines under different levels of NTC, where a specific percentage of videos (i.e., 30%, 60%, and 90%) are randomly shuffled. This setting is more challenging than the one adopted in NCR [16]. From the results, we draw the following observations:

First, our CURE not only consistently outperforms all strong baselines under low noise, but also maintains superior robustness under extreme high noise (90%). Notably, CURE achieves an average improvement of 17.11% over the most competitive baseline on the WebCVR dataset with a noisy rate of 60%. This substantial gain underscores the effectiveness of our evidential cross-modal uncertainty calibration mechanism, which alleviates the impact of noisy triplets and enables more robust video retrieval. This improvement arises from the synergistic interplay of three core components within CURE.

Second, vanilla video retrieval methods generally exhibit inferior performance compared to CURE, highlighting their vulnerability to NTC, particularly under extremely high noise levels (e.g., 90%). This observation validates the critical importance of explicitly addressing the issue of NTC. Interestingly, certain vanilla methods occasionally achieve the second-best performance, likely due to their sophisticated architectures specifically tailored for the CVR task. These results suggest that, while still susceptible to noise, such methods can effectively leverage clean data with well-aligned modalities to capture meaningful cross-modal relationships.

Third, robust methods generally exhibit stronger robustness across varying noise levels, further demonstrating the benefits of explicitly addressing the issue of NTC. Notably, however, our CURE still outperforms existing robust models by a considerable margin. This performance gap can be attributed to two key limitations of prior approaches: (1) insufficient modeling of evidential cross-modal uncertainty for noisy triplets, and (2) inadequate uncertainty-guided calibration for achieving noise-robust cross-modal alignment. Additional results are available in Appendix B.

4.3 Ablation Studies

To evaluate each component in CURE, we conduct an ablation study under a 60% noise rate, with the results summarized in Table 2.

4.3.1 Effect of UMF. To evaluate the effectiveness of UMF in CURE, we design three variant models: (1) *w/o UMF*, which removes the entire filtration module; (2) *w/o $\mathcal{L}_e$* , which removes the least-squares loss from the filter; and (3) *w/o $\mathcal{L}_{kl}$* , which removes the KL regularization term from the filter. As shown in Table 2, removing the UMF module results in a performance drop, providing evidence for the effectiveness of our design in capturing and modeling cross-modal alignment uncertainty to dynamically identify noisy triplets. Furthermore, the performance degradation observed when omitting either the least-squares loss $\mathcal{L}_e$ or the KL divergence term $\mathcal{L}_{kl}$ confirms that both components are critical for modeling uncertainty in UMF, enabling effective partitioning of the training data into fine-grained subsets.

4.3.2 Effect of RSA. In this part, we conduct an ablation study to evaluate the contribution of the RSA module. We construct a variant model (*w/o RSA*) that removes the entire semantic alignment

Table 2: Ablation study of CURE under the 60% noise rate.

UMF		RSA		MRA		WebCVR			FineCVR		
$\mathcal{L}_e$	$\mathcal{L}_{kl}$	$\mathcal{L}_{RAL}$	$\mathcal{L}_{MRA}$			$R@1$	$R@5$	$R@10$	$R@1$	$R@5$	$R@10$
✓	✓	✓	✓			40.39	68.29	77.79	32.95	66.53	79.78
		✓	✓			40.31	67.32	77.39	32.64	65.87	79.56
✓		✓	✓			40.32	67.36	77.63	32.42	65.70	79.02
	✓	✓	✓			40.15	67.16	77.16	31.41	64.80	77.92
✓	✓		✓			34.44	61.79	71.99	31.19	64.31	77.70
✓	✓	✓				38.45	65.26	74.89	31.87	64.86	78.10

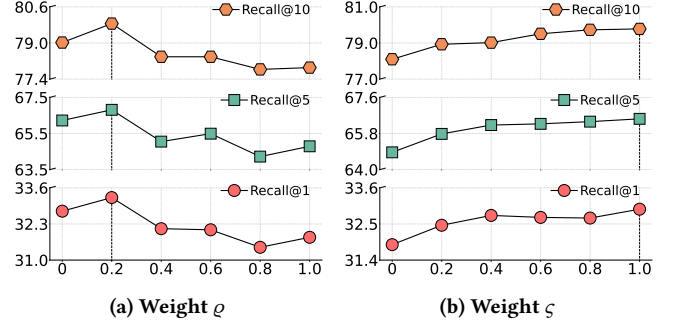


Figure 3: Sensitivity analysis of weight ρ and ζ on the FineCVR dataset.

component. Table 2 shows that removing this module results in a substantial performance drop, validating the importance of our design in rectifying soft margins and relaxing the optimization over all negatives during the cross-modal alignment process. The RSA module assigns higher margins to matched triplets with low uncertainty and lower margins to mismatched triplets with high uncertainty. Moreover, by incorporating all negative triplets rather than relying solely on the hardest ones, it enables more stable and comprehensive training. With this design, CURE achieves noise-robust cross-modal alignment in the CVR task.

4.3.3 Effect of MRA. To analyze the impact of the MRA module, we construct a variant model (*w/o MRA*) that removes the entire relation augmentation component. As shown in Table 2, removing this module results in a significant performance decline, underscoring its critical role in mining discriminative semantic information and mitigating the adverse effects of partial matching within the clean set. By leveraging auxiliary video captions, the MRA module enriches the semantic alignment process and facilitates multi-level cross-modal alignment, thereby enhancing robustness against NTC.

4.4 Parameter Sensitivity

Figure 3 presents the sensitivity analysis of CURE's parameters.

4.4.1 Weight ρ . Figure 3(a) illustrates the effect of the weight ρ for the loss function $\mathcal{L}_{UMF}$. The performance of CURE initially improves as ρ increases, reaches a peak, and then declines as ρ becomes too large. We observe that $\rho = 0.2$ yields optimal performance under the 60% noise setting. This result validates the efficacy of our design in the UMF module for identifying noisy triplets.

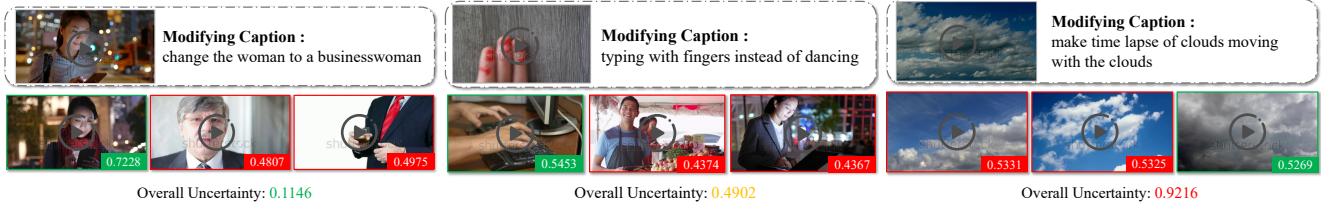


Figure 4: Retrieved examples on the WebCVR under 60% noise. The first row shows the composed query, consisting of a reference video and a modifying caption. The second row displays the top-3 retrieved videos. Correctly videos are highlighted with green boxes, while incorrectly ones are outlined in red. Estimated uncertainty is shown in bold brackets in captions, while similarity scores appear as white overlays on video frames, serving as sub-captions and exemplar annotations, respectively.

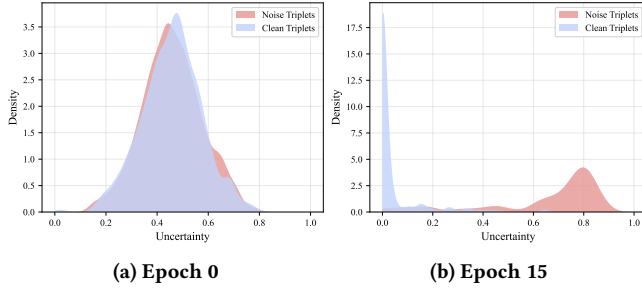


Figure 5: Visualization of the uncertainty distribution for clean and noisy triplets at different training stages of our CURE, conducted on the WebCVR dataset under a 60% noise.

4.4.2 Weight ζ . Figure 3 (b) demonstrates that extracting multi-level discriminative semantic information and reconciling multi-level text–video similarity distributions for cross-modal alignment at both the video and caption levels generally improves model performance. Specifically, the performance of CURE increases as the weight ζ for the loss function $\mathcal{L}_{TRA}$ is increased, achieving optimal results at $\zeta = 1.0$. Additional results are available in Appendix C.

4.5 Further Analysis

• **Retrieval Quality.** To illustrate the effectiveness of CURE, we present a qualitative analysis by randomly selecting three examples on the WebCVR test set. Unlike prior work, CURE not only retrieves videos based on the modifying caption but also estimates the uncertainty associated with each retrieved result. In Figure 4, we observe that the overall uncertainty exhibits a negative correlation with retrieval performance. Specifically, the amount and ranking of correctly retrieved results will impact the score of uncertainty. The more and higher up the correct retrieved results are, the lower the uncertainty is, and vice versa. In addition, high similarity does not necessarily imply a correct match, but uncertainty offers a more reliable evaluation, i.e., an incorrect retrieval with high similarity typically also exhibits high uncertainty. It shows the effectiveness of our uncertainty-guided division strategy by explicitly quantifying retrieval uncertainty, thereby enhancing the interpretability of the CVR task. Additional results are available in Appendix D.

• **Uncertainty Visualization.** We conduct experiments under a 60% noise setting on the WebCVR dataset to visualize the evolution of uncertainty and compare the cross-modal alignment uncertainty captured by CURE. As shown in Figure 5, the uncertainty distribution exhibits a polarizing trend throughout training. Specifically, the uncertainty associated with clean triplets progressively decreases,

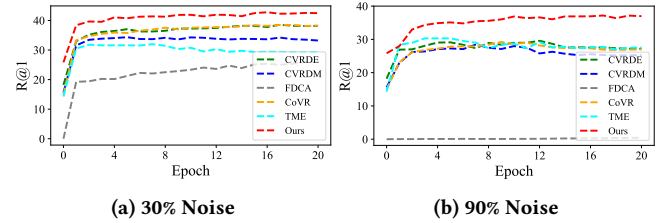


Figure 6: Test performance (R1) across training epochs on the WebCVR dataset under 30% and 90% noise settings.

while that of noisy triplets increases. This shift in distribution highlights the effectiveness of our CURE in assigning lower penalties to clean pairs and higher penalties to noisy ones. These results further validate our design choice: uncertainty can be naturally leveraged to distinguish clean from noisy triplets, thereby enhancing robustness against NTC in the CVR task.

• **Model Robustness.** To further analyze the robustness of our CURE, we present visualization results from the cross-modal training process, comparing CURE against four video-centric baselines: CoVR, CVRDE, CVRDM, and FDCA. As illustrated in Figure 6, CURE not only demonstrates strong performance under noisy conditions but also effectively alleviates the adverse effects of NTC.

5 Conclusion

We addressed the under-explored problem of noisy triplet correspondence in composed video retrieval. Our solution, CURE, brings evidential thinking into cross-modal learning: UMF learns uncertainty by parameterizing bidirectional evidence and adaptively assigning weights to training triplets, enabling the identification of noisy data instances. RSA rectifies the calibrated soft margins and relaxes optimization over all negatives, thereby achieving noise-robust alignment. MRA mines multi-level discriminative information to facilitate alignment at both the video and caption levels, further mitigating the adverse effects of noise. Extensive experiments on two datasets demonstrate the robustness and superiority of CURE. Future work will explore additional uncertainty techniques to improve retrieval performance.

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Table 3: Statistics of three datasets.

Dataset	# Triplets	# Visuals	# words	# Train	# Test
WebCVR	1,651,345	133,219	21,098	123,942	3,000
FineCVR	1,010,071	136,547	20,961	109,541	6,364
WebCVR-8M	2,556	2,444	1,935	–	2,556

A Experimental Settings

- *Datasets.* For evaluation, we use two benchmark datasets following prior works [36, 40]: **WebCVR** [36] is constructed based on the WebVid2M dataset [3]. The training split comprises 131K unique videos and 467K distinct change descriptions. Each video is, on average, associated with 12.7 triplets, and the mean length of a change description is 4.8 words. The test set contains 3.2K triplets that have been manually curated to ensure high quality and consistency for evaluation. **FineCVR** dataset [40] contains 1,000,028 triplets for training and 10,043 triplets for testing. It is designed to support fine-grained video retrieval by incorporating rich visual semantics, including actions, scenes, objects, and attributes. In addition, **WebCVR-8M** dataset contains 2.5k manually annotated high-quality triplets that are not included in WebCVR, making it well-suited for assessing generalization capabilities.

- *Baselines.* To evaluate the effectiveness of our CURE, we compare CURE with 13 competitive baselines divided into three groups: (1) *Fusion-based methods*: CLIP [30], BLIP [23], CoVR [36], FDCA [40], CVRDE [35] and CVRDM [34]. (2) *Inversion-based methods*: Pic2word [31], I2W [33], and LinCIR [10]. (3) *Robust methods*: Uncer-R [6], NCR [16], RCL [14], and TME [24].

Specifically: (1) CLIP freezes its ViT and text encoder, training an MLP to fuse reference video and modification text. (2) BLIP fine-tunes an MLP atop the pretrained BLIP model for video-text retrieval. (3) CoVR freezes the ViT and fine-tunes the image-grounded text encoder. (4) FDCA disentangles textual features at sentence and token levels for fine-grained semantics. (5) CVRDE leverages detailed language to encode query-specific context. (6) CVRDM enhances retrieval via cross-attention fusion with a grounded text encoder. (7) Pic2Word uses zero-shot textual inversion with contrastive loss to produce pseudo-word tokens. (8) I2W introduces Context-I2W with an Intent View Selector and Visual Target Extractor. (9) LinCIR trains using language-only data, projecting text into token embeddings. (10) Uncer-R models uncertainty and applies regularization for retrieval. (11) NCR rectifies noisy correspondence through neural memorization and adaptive co-teaching. (12) RCL applies contrastive learning to estimate matching probabilities and reduce over/underfitting. (13) TME mitigates noisy correspondence via RFQ, PTE, and TOP modules.

- *Implementation.* For caption extraction of reference videos, we adopt the pre-trained BLIP-2 model with the OPT-6.7B language model as the captioning backbone. The retriever is built upon the BLIP-Large model, where the ViT component is kept frozen, while the remaining modules are fine-tuned. We train the model using the Adam optimizer with an initial learning rate of $1e-5$ and a batch size of 128. The key hyperparameters are configured as follows: the initial margin γ in the RAL module is set to 0.1, the threshold η for dividing the training data is set to 0.5, and the temperature

parameter τ is set to 0.015. Our CURE is implemented in PyTorch and trained on two NVIDIA RTX 4090 GPUs.

B Additional Main Results

As illustrated in Table 4, we conduct comparative experiments under the 0% noise setting. We obtain consistent findings that corroborate the observations presented in Section 4.2. Specifically, our proposed CURE consistently outperforms all strong baselines, further validating its effectiveness. Although the three modules in CURE are primarily designed to address the NTC problem, they remain effective even on clean triplets. This is because visual variations are often too complex to be fully captured by textual modifications, thereby leading to implicit instances of NTC.

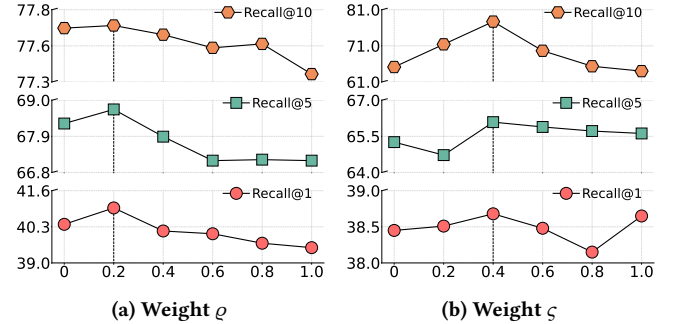


Figure 7: Sensitivity analysis of weight ρ and ζ on the WebCVR dataset.

C Additional Parameter Analysis

Loss weights ρ and ζ control the contributions of $\mathcal{L}_{UMF}$ and $\mathcal{L}_{MRA}$, respectively. As shown in Figure 7, we analyze the impact of varying ρ and ζ over the range $\{0.0, 0.2, 0.4, 0.6, 0.8, 1.0\}$ on the WebCVR dataset. Experimental results reveal that setting $\rho = 0.2$ and $\zeta = 0.4$ yields the best performance. These findings underscore the importance of our evidential cross-modal uncertainty calibration mechanism in addressing the NTC problem. Moreover, the results further validate the effectiveness of our design under the uncertainty-guided learning paradigm.

D Retrieval Quality

We present a qualitative analysis by randomly selecting three retrieved examples on the FineCVR test set. In Figure 8, we observe that the overall uncertainty exhibits a negative correlation with retrieval performance. Specifically, lower uncertainty is associated with more accurate retrievals, where the matched results typically show a high degree of similarity. In contrast, higher uncertainty often corresponds to mismatched results, which exhibit significantly lower semantic similarity. Furthermore, we observe that leveraging deterministic similarity is insufficient to reliably reflect retrieval confidence. In some cases, high similarity scores correspond to factually mismatched pairs, while low similarity scores may still result in correct retrievals. This observation validates the effectiveness of our uncertainty-guided division strategy by explicitly quantifying retrieval uncertainty.

Table 4: Performance comparison on two real-world datasets under varying noise rates. The best results are in bold font and the second underlined. Lower values of MnR, and higher values of R@k, indicate better performance.

Dataset	Noise	Metric	BLIP	CLIP	CoVR	CVRDE	CVRDM	FDCA	Pic2word	I2W	LinCIR	Uncer-R	NCR	RCL	TME	CURE	Improv.
WebCVR	0%	R@1	18.37	18.23	36.33	<u>38.07</u>	34.47	28.50	27.57	26.21	29.53	12.87	16.00	16.70	30.31	41.97	10.24%
		R@5	39.10	42.00	60.43	<u>63.90</u>	60.57	55.70	49.00	48.32	51.60	25.83	35.90	35.37	55.41	70.47	10.28%
		R@10	50.57	52.17	72.57	<u>73.53</u>	70.43	65.27	57.47	58.35	61.00	33.10	45.32	44.97	64.55	80.00	8.80%
		MnR	73.90	60.75	36.13	<u>29.91</u>	43.18	47.92	97.25	73.63	69.72	155.09	95.27	126.29	51.10	17.12	42.76%
FineCVR	0%	R@1	19.12	18.53	32.87	32.18	<u>33.53</u>	20.45	7.87	7.43	9.73	5.11	5.01	7.34	27.60	34.46	2.77%
		R@5	46.42	46.65	66.58	66.48	<u>67.27</u>	50.78	22.96	21.34	27.55	17.88	19.96	22.85	59.04	68.54	1.89%
		R@10	60.67	60.61	79.71	79.34	<u>80.01</u>	65.25	32.57	30.48	38.67	26.65	28.93	32.35	72.71	81.13	1.40%
		MnR	35.10	29.57	12.08	<u>11.69</u>	11.98	22.71	197.04	206.65	162.98	210.00	162.21	202.22	24.29	10.60	9.32%

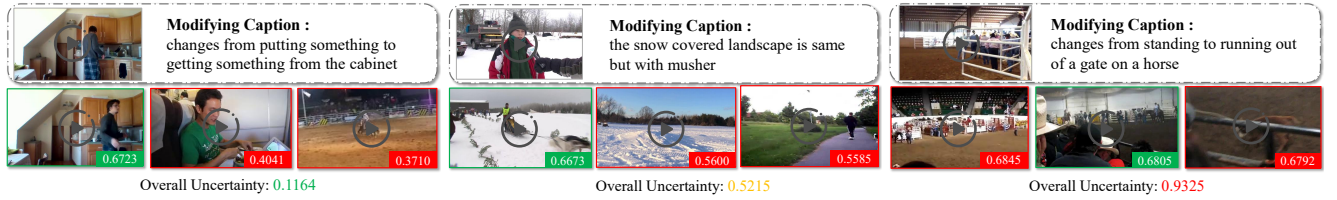
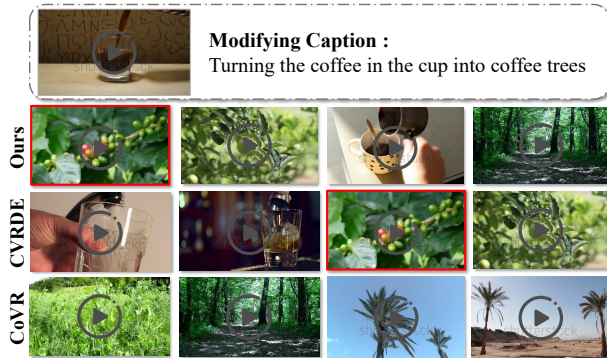
**Figure 8: Retrieved examples on the FineCVR under 60% noise. The first row shows the composed query, consisting of a reference video and a modifying caption. The second row displays the top-3 retrieved videos. Correctly videos are highlighted with green boxes, while incorrectly ones are outlined in red. Estimated uncertainty is shown in bold brackets in captions, while similarity scores appear as white overlays on video frames, serving as sub-captions and exemplar annotations, respectively.****Figure 9: Qualitative analysis of two CVR baselines with respect to the R@5 metric on the WebCVR dataset. The target video is highlighted with red outlines.**

Figure 9 visualizes one successful retrieval example, where the textual instructions affect different semantic aspects of the reference videos. Compared to a strong baseline (e.g., CoVR and CVRDE), we observe that CoVR often fails to attend to the relevant visual cues described in the modification text. For instance, in Case One, it overlooks key concepts such as “cup” and “coffee.” This highlights the limitation of rigid retrieval strategies, which struggle to adapt to complex modifications involving multiple objects and fine-grained changes. In contrast, our CURE effectively captures all intended modifications and accurately retrieves the correct target videos. This improvement is attributed to our evidential cross-modal uncertainty calibration, which enhances the robustness of CVR by quantifying the uncertainty arising from NTC.